

Theory of Commutative Algebra 6 Theory Notes

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This file is a professionally reorganized and cleaned version.

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Chapter 1

Examples

1.1 Examples and intuition for $D(A)$

Recall that

$$D(A) = \{\mathfrak{p} \in \text{Spec}(A) : \exists a \in A \text{ such that } \mathfrak{p} \text{ is minimal in } V(\text{Ann}(a))\}.$$

Thus $D(A)$ is built from annihilators of individual elements. It records the prime ideals that appear as the smallest prime-level reasons why some element is killed by multiplication. Equivalently, it is a prime-theoretic description of the set of zero divisors.

A useful slogan is:

$$\text{zero divisors} = \bigcup_{\mathfrak{p} \in D(A)} \mathfrak{p}.$$

1.1.1 1. Domain case: $A = k[t]$

Let $A = k[t]$, where k is a field. Since A is an integral domain, the only zero divisor is 0. For every nonzero $a \in A$,

$$\text{Ann}(a) = 0.$$

Hence the only prime that can arise is

$$D(A) = \{(0)\}.$$

Thus part (a) becomes

$$x \text{ is a zero divisor} \iff x \in (0),$$

which says exactly that only 0 is a zero divisor in a domain.

1.1.2 2. Two crossing components: $A = k[x, y]/(xy)$

Let

$$A = k[x, y]/(xy),$$

and write $\bar{x}, \bar{y}$ for the classes of x, y in A . Since

$$\bar{x}\bar{y} = 0,$$

both $\bar{x}$ and $\bar{y}$ are zero divisors.

Moreover,

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}).$$

Therefore

$$(\bar{x}), (\bar{y}) \in D(A).$$

In fact these are exactly the minimal primes of A , so

$$D(A) = \{(\bar{x}), (\bar{y})\}.$$

Hence the set of zero divisors is

$$(\bar{x}) \cup (\bar{y}).$$

Geometric interpretation. The scheme $\text{Spec}(A)$ is the union of the two coordinate axes crossing at the origin. The primes in $D(A)$ correspond to the irreducible components. An element is a zero divisor precisely when it vanishes on at least one whole component.

1.1.3 3. Nilpotent example: $A = k[t]/(t^2)$

Let

$$A = k[t]/(t^2),$$

and let $\bar{t}$ denote the class of t . Then

$$\bar{t}^2 = 0,$$

so $\bar{t}$ is nilpotent and hence a zero divisor.

Also

$$\text{Ann}(\bar{t}) = (\bar{t}).$$

Therefore

$$D(A) = \{(\bar{t})\}.$$

This example shows that $D(A)$ detects zero divisors coming from nilpotent structure, not only from reducible geometry.

Interpretation. Here $\text{Spec}(A)$ has only one point, but the ring is nonreduced. So $D(A)$ can detect infinitesimal thickening as well as decomposition into components.

1.1.4 4. Localization removes primes that meet S

Again let

$$A = k[x, y]/(xy), \quad S = \{1, \bar{x}, \bar{x}^2, \dots\}.$$

In $S^{-1}A$, the element $\bar{x}$ becomes invertible. Since

$$\bar{x}\bar{y} = 0,$$

it follows that

$$\bar{y} = 0$$

in $S^{-1}A$. Thus

$$S^{-1}A \cong k[\bar{x}, \bar{x}^{-1}],$$

which is a domain.

Hence

$$D(S^{-1}A) = \{(0)\}$$

inside $\text{Spec}(S^{-1}A)$.

Under the canonical identification

$$\text{Spec}(S^{-1}A) \cong \{\mathfrak{p} \in \text{Spec}(A) \mid \mathfrak{p} \cap S = \emptyset\},$$

this prime corresponds to $(\bar{y})$. Therefore, viewed as a subset of $\text{Spec}(A)$,

$$D(S^{-1}A) = \{(\bar{y})\}.$$

But

$$D(A) = \{(\bar{x}), (\bar{y})\},$$

and only $(\bar{y})$ is disjoint from S . Hence

$$D(S^{-1}A) = D(A) \cap \text{Spec}(S^{-1}A).$$

Interpretation. Localization keeps only the primes disjoint from S . Any prime in $D(A)$ that meets S disappears after localization.

1.1.5 5. Embedded associated prime: $A = k[x, y]/(x^2, xy)$

Let

$$R = k[x, y], \quad I = (x^2, xy), \quad A = R/I.$$

Since

$$I = x(x, y) = (x) \cap (x^2, y),$$

we obtain a primary decomposition of I . Therefore the associated primes of A are

$$\text{Ass}(A) = \{(x)/I, (x, y)/I\}.$$

Now let us see that both belong to $D(A)$.

First, $(x)/I$ is minimal over

$$\text{Ann}_A(1) = 0,$$

so

$$(x)/I \in D(A).$$

Second, the class $\bar{x} \in A$ satisfies

$$x^2 = 0, \quad xy = 0.$$

Hence

$$\text{Ann}_A(\bar{x}) = (\bar{x}, \bar{y}) = (x, y)/I,$$

so

$$(x, y)/I \in D(A).$$

Thus

$$D(A) = \{(x)/I, (x, y)/I\} = \text{Ass}(A).$$

Why this matters. The prime $(x, y)/I$ is an embedded prime, not a minimal prime. So $D(A)$ can detect hidden nonreduced structure inside a component.

1.1.6 6. Product ring example: $A = k \times k$

Let

$$A = k \times k.$$

Then

$$(1, 0) \cdot (0, 1) = 0,$$

so both $(1, 0)$ and $(0, 1)$ are zero divisors.

The prime ideals are

$$\mathfrak{p}_1 = 0 \times k, \quad \mathfrak{p}_2 = k \times 0.$$

Also

$$\text{Ann}(1, 0) = 0 \times k = \mathfrak{p}_1, \quad \text{Ann}(0, 1) = k \times 0 = \mathfrak{p}_2.$$

Therefore

$$D(A) = \{\mathfrak{p}_1, \mathfrak{p}_2\}.$$

An element $(a, b) \in k \times k$ is a zero divisor if and only if

$$a = 0 \quad \text{or} \quad b = 0.$$

So again

$$\{\text{zero divisors}\} = \mathfrak{p}_1 \cup \mathfrak{p}_2.$$

This is the most concrete algebraic model of a reducible space with two disjoint components.

1.1.7 7. Many zero divisors, one prime

Let

$$A = k[x, y]/(x^2, xy, y^2).$$

Then the maximal ideal

$$\mathfrak{m} = (\bar{x}, \bar{y})$$

satisfies

$$\mathfrak{m}^2 = 0.$$

Hence every element of $\mathfrak{m}$ is nilpotent and therefore a zero divisor.

However, $\text{Spec}(A)$ has only one prime ideal, namely $\mathfrak{m}$. Thus

$$D(A) = \{\mathfrak{m}\}.$$

This example shows that many different zero divisors may all be controlled by the same prime in $D(A)$.

1.1.8 8. Relation with $\text{Ass}(A)$

If the zero ideal of A has a primary decomposition, then part (c) says that

$$D(A) = \text{Ass}(A).$$

So in Noetherian rings with primary decomposition, $D(A)$ is exactly the set of associated primes of the A -module A .

In particular:

- in reduced rings such as $k[x, y]/(xy)$, $D(A)$ often consists of minimal primes;
- in nonreduced rings such as $k[x, y]/(x^2, xy)$, $D(A)$ may also contain embedded primes.

1.1.9 9. Relation with zero divisors

Part (a) gives the important identity

$$\{\text{zero divisors of } A\} = \bigcup_{\mathfrak{p} \in D(A)} \mathfrak{p}.$$

Thus $D(A)$ is a prime-theoretic description of the zero divisors. When primary decomposition exists, this becomes

$$\{\text{zero divisors of } A\} = \bigcup_{\mathfrak{p} \in \text{Ass}(A)} \mathfrak{p}.$$

1.1.10 10. How to compute $D(A)$ in practice

There are two standard methods.

Method 1: via annihilators of elements. Choose simple elements $a \in A$, compute $\text{Ann}(a)$, and find the minimal primes over $\text{Ann}(a)$. All primes obtained this way belong to $D(A)$.

Method 2: via primary decomposition. If A is Noetherian and 0 has a primary decomposition, compute that decomposition and take the radicals of its primary components. These radicals are exactly the primes in $D(A)$, because then

$$D(A) = \text{Ass}(A).$$

1.1.11 11. Final summary

The most important examples to remember are:

$$k[t], \quad k[x, y]/(xy), \quad k[t]/(t^2), \quad k[x, y]/(x^2, xy), \quad k \times k.$$

They illustrate, respectively:

- the domain case;
- reducible geometry with two components;
- nilpotent structure;
- embedded associated primes;
- and a completely explicit decomposed ring.

Together they show that $D(A)$ measures where zero divisors come from, how those primes behave under localization, and why in the Noetherian case $D(A)$ agrees with the set of associated primes.

1.2 Concrete polynomial and ring examples for $D(A)$

Recall that

$$D(A) = \{\mathfrak{p} \in \operatorname{Spec}(A) : \exists a \in A \text{ such that } \mathfrak{p} \text{ is minimal in } V(\operatorname{Ann}(a))\}.$$

Part (a) says that

$$\{\text{zero divisors of } A\} = \bigcup_{\mathfrak{p} \in D(A)} \mathfrak{p}.$$

We now present this on explicit rings and explicit polynomial quotients.

1.2.1 1. Arithmetic example: $A = \mathbb{Z}/6\mathbb{Z}$

The elements are

$$\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}.$$

The zero divisors are

$$\bar{0}, \bar{2}, \bar{3}, \bar{4},$$

because

$$\bar{2} \cdot \bar{3} = \bar{0}, \quad \bar{3} \cdot \bar{4} = \bar{0}.$$

Now compute annihilators:

$$\operatorname{Ann}(\bar{2}) = \{\bar{r} \in \mathbb{Z}/6\mathbb{Z} : \bar{r}\bar{2} = \bar{0}\} = \{\bar{0}, \bar{3}\} = (\bar{3}),$$

and

$$\operatorname{Ann}(\bar{3}) = \{\bar{0}, \bar{2}, \bar{4}\} = (\bar{2}).$$

The prime ideals of $\mathbb{Z}/6\mathbb{Z}$ are

$$(\bar{2}), \quad (\bar{3}),$$

so

$$D(A) = \{(\bar{2}), (\bar{3})\}.$$

Thus

$$\{\text{zero divisors}\} = (\bar{2}) \cup (\bar{3}).$$

1.2.2 2. One-variable polynomial example: $A = k[x]/(x(x-1))$

Let $\bar{x}$ denote the class of x in A . Then

$$\bar{x}(\bar{x} - 1) = 0,$$

so both $\bar{x}$ and $\bar{x} - 1$ are zero divisors.

We now compute their annihilators.

Since $(\bar{x} - 1)\bar{x} = 0$, every multiple of $\bar{x} - 1$ annihilates $\bar{x}$. In fact,

$$\text{Ann}(\bar{x}) = (\bar{x} - 1).$$

Similarly,

$$\text{Ann}(\bar{x} - 1) = (\bar{x}).$$

Hence

$$(\bar{x}), (\bar{x} - 1) \in D(A),$$

and in fact

$$D(A) = \{(\bar{x}), (\bar{x} - 1)\}.$$

So the zero divisors are precisely the elements lying in one of these two prime ideals.

Examples.

$\bar{x}$ is a zero divisor, $\bar{x} - 1$ is a zero divisor, $\bar{1}$ is not a zero divisor.

Geometric meaning. The equation

$$x(x-1) = 0$$

describes the union of the two points $x = 0$ and $x = 1$ on the affine line.

1.2.3 3. Two-variable geometric example: $A = k[x, y]/(xy)$

Let $\bar{x}, \bar{y}$ be the classes of x, y in A . Then

$$\bar{x}\bar{y} = 0,$$

so $\bar{x}$ and $\bar{y}$ are zero divisors.

To compute $\text{Ann}(\bar{x})$, we look for all $\bar{f} \in A$ such that

$$\bar{f}\bar{x} = 0.$$

This means

$$fx \in (xy)$$

in $k[x, y]$. Since fx must be divisible by xy , the polynomial f must be divisible by y . Therefore

$$\text{Ann}(\bar{x}) = (\bar{y}).$$

Similarly,

$$\text{Ann}(\bar{y}) = (\bar{x}).$$

Hence

$$D(A) = \{(\bar{x}), (\bar{y})\}.$$

So the zero divisors are exactly the elements in

$$(\bar{x}) \cup (\bar{y}).$$

Concrete classes. Take

$$a = \bar{x}, \quad b = \bar{y}, \quad c = \bar{x} + \bar{y}, \quad d = \bar{x} + 1.$$

Then:

- a is a zero divisor since $a\bar{y} = 0$;
- b is a zero divisor since $b\bar{x} = 0$;
- $c \notin (\bar{x}) \cup (\bar{y})$, so c is not a zero divisor;
- $d \notin (\bar{x}) \cup (\bar{y})$, so d is not a zero divisor.

Geometric meaning. The variety

$$V(xy) = V(x) \cup V(y)$$

is the union of the two coordinate axes. The primes in $D(A)$ correspond to those two components.

1.2.4 4. Nilpotent polynomial example: $A = k[t]/(t^2)$

Let $\bar{t}$ be the class of t in A . Then

$$\bar{t}^2 = 0,$$

so $\bar{t}$ is nilpotent and hence a zero divisor.

Now $\bar{f} \in \text{Ann}(\bar{t})$ if and only if

$$\bar{f}\bar{t} = 0,$$

that is, if and only if

$$ft \in (t^2).$$

This holds exactly when $f \in (t)$. Hence

$$\text{Ann}(\bar{t}) = (\bar{t}),$$

and therefore

$$D(A) = \{(\bar{t})\}.$$

Every element of A has the form

$$a + b\bar{t}, \quad a, b \in k.$$

If $a \neq 0$, then $a + b\bar{t}$ is a unit. If $a = 0$, then the element equals $b\bar{t}$, which is nilpotent. Thus the zero divisors are exactly the ideal

$$(\bar{t}).$$

1.2.5 5. Embedded-prime example: $A = k[x, y]/(x^2, xy)$

Let

$$A = k[x, y]/(x^2, xy),$$

and let $\bar{x}, \bar{y}$ be the classes of x, y . Then in A ,

$$\bar{x}^2 = 0, \quad \bar{x}\bar{y} = 0.$$

So $\bar{x}$ is annihilated by both $\bar{x}$ and $\bar{y}$, and therefore

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}).$$

Hence

$$(\bar{x}, \bar{y}) \in D(A).$$

On the other hand,

$$(x^2, xy) = x(x, y) = (x) \cap (x^2, y),$$

so the associated primes are

$$(\bar{x}), \quad (\bar{x}, \bar{y}).$$

Therefore

$$D(A) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

Interpretation. The prime $(\bar{x})$ is the minimal prime, while $(\bar{x}, \bar{y})$ is an embedded prime. This shows that $D(A)$ can detect not only visible components but also hidden nilpotent structure inside a component.

1.2.6 6. Concrete localization example

Take again

$$A = k[x, y]/(xy),$$

and let

$$S = \{1, \bar{x}, \bar{x}^2, \bar{x}^3, \dots\}.$$

In the localization $S^{-1}A$, the element $\bar{x}$ becomes invertible. Since

$$\bar{x}\bar{y} = 0,$$

we may multiply by $\bar{x}^{-1}$ and obtain

$$\bar{y} = 0.$$

Hence

$$S^{-1}A \cong k[\bar{x}, \bar{x}^{-1}],$$

which is a domain.

Before localization,

$$D(A) = \{(\bar{x}), (\bar{y})\}.$$

After localization, the prime $(\bar{x})$ disappears because it meets S , while $(\bar{y})$ survives. Thus

$$D(S^{-1}A) = \{(\bar{y})\}$$

when viewed inside $\text{Spec}(A)$.

This is the concrete meaning of the formula

$$D(S^{-1}A) = D(A) \cap \text{Spec}(S^{-1}A).$$

1.2.7 7. Practical computation rule

For a quotient ring

$$A = k[x_1, \dots, x_n]/I,$$

one can compute $D(A)$ concretely by the following steps:

1. Find obvious zero divisors, that is, classes whose product is zero modulo I .
2. For a chosen element $a \in A$, compute $\text{Ann}(a)$ by solving

$$fa \in I.$$

3. Find the minimal primes over $\text{Ann}(a)$.
4. Collect all primes that arise.

If 0 has a primary decomposition, then one may instead compute $\text{Ass}(A)$, because in that case

$$D(A) = \text{Ass}(A).$$

1.2.8 8. Final summary

The most concrete examples to remember are:

$$\mathbb{Z}/6\mathbb{Z}, \quad k[x]/(x(x-1)), \quad k[x, y]/(xy), \quad k[t]/(t^2), \quad k[x, y]/(x^2, xy).$$

They illustrate:

- arithmetic zero divisors;
- decomposition into two points;
- decomposition into two axes;
- nilpotent zero divisors;
- and embedded associated primes.

1.3 Classes in quotient rings, minimal primes, and associated primes

1.3.1 1. What do $\bar{x}, \bar{y}$ mean?

Let

$$A = R/I$$

be a quotient ring. For any element $f \in R$, its image in A is denoted by

$$\bar{f} \quad \text{or sometimes} \quad [f].$$

Thus

$$\bar{f} = f + I.$$

So:

$$\bar{x} = x + I, \quad \bar{y} = y + I, \quad \overline{x + y} = x + y + I.$$

This is the residue class (or coset) of f modulo I .

Two elements $f, g \in R$ define the same class in A if and only if

$$\bar{f} = \bar{g} \iff f - g \in I.$$

1.3.2 2. Concrete examples of classes

Example: $A = k[x, y]/(xy)$

Here $I = (xy)$. Then:

$$\bar{x} = x + (xy), \quad \bar{y} = y + (xy).$$

Also

$$\overline{x + xy} = \bar{x}$$

because

$$(x + xy) - x = xy \in (xy).$$

Moreover,

$$\overline{xy} = 0, \quad \overline{x^2y} = 0,$$

since $xy, x^2y \in (xy)$.

However,

$$\bar{x} \neq 0, \quad \bar{y} \neq 0,$$

because $x, y \notin (xy)$.

Thus in this quotient ring,

$$\bar{x}\bar{y} = 0$$

while neither $\bar{x}$ nor $\bar{y}$ is zero.

Example: $A = k[t]/(t^2)$

Here

$$\bar{t} \neq 0$$

but

$$\bar{t}^2 = 0.$$

Every element has the form

$$a + b\bar{t}, \quad a, b \in k.$$

Example: $A = \mathbb{Z}/6\mathbb{Z}$

The class $\bar{2}$ means all integers congruent to 2 modulo 6:

$$\bar{2} = \{\dots, -10, -4, 2, 8, 14, \dots\}.$$

Also

$$\bar{2}\bar{3} = \bar{0}.$$

1.3.3 3. How to think about quotient classes

The quotient ring R/I is obtained from R by imposing all relations coming from I .

For example:

$k[x, y]/(xy)$ means that in the quotient we force $xy = 0$;

$k[x, y]/(x^2, xy)$ means that in the quotient we force $x^2 = 0$ and $xy = 0$;

$k[x]/(x(x-1))$ means that in the quotient we force $x(x-1) = 0$.

1.3.4 4. Minimal primes

A prime ideal $\mathfrak{p} \subset A$ is called a *minimal prime* if it is minimal among the prime ideals of A .

If

$$A = R/I,$$

then minimal primes of A correspond to prime ideals of R that are minimal over I .

We write

$$\text{Min}(A)$$

for the set of minimal primes.

Interpretation. Minimal primes describe the top-level irreducible components of the ring or affine scheme.

Example: $A = k[x]$

Since A is a domain, (0) is prime and

$$\text{Min}(A) = \{(0)\}.$$

Example: $A = k[x, y]/(xy)$

The ideal (xy) has two minimal primes:

$$(x), \quad (y).$$

Hence in the quotient,

$$\text{Min}(A) = \{(\bar{x}), (\bar{y})\}.$$

Example: $A = k[x, y]/(x^2)$

The only prime minimal over (x^2) is (x) . Hence

$$\text{Min}(A) = \{(\bar{x})\}.$$

Example: $A = k[x, y]/(x^2, xy)$

Since

$$\sqrt{(x^2, xy)} = (x),$$

the only minimal prime is

$$\text{Min}(A) = \{(\bar{x})\}.$$

1.3.5 5. Associated primes

A prime ideal $\mathfrak{p} \subset A$ is an *associated prime* of the A -module A if

$$\mathfrak{p} = \text{Ann}_A(a)$$

for some element $a \in A$.

We write

$$\text{Ass}(A)$$

for the set of associated primes of A .

Interpretation. Associated primes describe primes that arise as actual annihilators of elements. They detect zero-divisor structure and possible embedded components.

Example: $A = k[x, y]/(xy)$

We have

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}).$$

Therefore

$$\text{Ass}(A) = \{(\bar{x}), (\bar{y})\}.$$

Here

$$\text{Ass}(A) = \text{Min}(A).$$

Example: $A = k[t]/(t^2)$

We have

$$\text{Ann}(\bar{t}) = (\bar{t}).$$

Hence

$$\text{Ass}(A) = \{(\bar{t})\} = \text{Min}(A).$$

Example: $A = k[x, y]/(x^2, xy)$

First,

$$\text{Min}(A) = \{(\bar{x})\}.$$

Now in A ,

$$\bar{x}^2 = 0, \quad \bar{x}\bar{y} = 0.$$

Hence

$$(\bar{x}, \bar{y}) \subseteq \text{Ann}(\bar{x}).$$

Conversely, if $f\bar{x} = 0$, then

$$fx \in (x^2, xy) = x(x, y),$$

so $f \in (x, y)$. Therefore

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}).$$

Thus

$$(\bar{x}, \bar{y}) \in \text{Ass}(A).$$

Since $(\bar{x})$ is the minimal prime, it is also associated in this Noetherian case. Hence

$$\text{Ass}(A) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

So

$$\text{Min}(A) = \{(\bar{x})\} \subsetneq \text{Ass}(A) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

The prime $(\bar{x}, \bar{y})$ is an embedded associated prime.

1.3.6 6. Comparison

Minimal primes describe the irreducible components of the ring:

geometry at the top level.

Associated primes describe annihilators of actual elements:

zero-divisor and embedded structure.

Thus in general,

$$\text{Min}(A) \subseteq \text{Ass}(A),$$

and the inclusion may be strict.

1.3.7 7. Summary table

A	$\text{Min}(A)$	$\text{Ass}(A)$
$k[x]$	$\{(0)\}$	$\{(0)\}$
$k[x, y]/(xy)$	$\{(\bar{x}), (\bar{y})\}$	$\{(\bar{x}), (\bar{y})\}$
$k[t]/(t^2)$	$\{(\bar{t})\}$	$\{(\bar{t})\}$
$k[x, y]/(x^2, xy)$	$\{(\bar{x})\}$	$\{(\bar{x}), (\bar{x}, \bar{y})\}$

1.3.8 8. Relation with $D(A)$

In the problem, $D(A)$ is defined via primes minimal over annihilators of elements. In the Noetherian case with primary decomposition,

$$D(A) = \text{Ass}(A).$$

Moreover, the set of zero divisors is

$$\bigcup_{\mathfrak{p} \in D(A)} \mathfrak{p}.$$

So associated primes are the natural language for understanding zero divisors.

1.4 Concrete meaning of x , x^2 , $\mathbb{Z}/p\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$, quotient classes, minimal primes, and associated primes

1.4.1 1. What is x in a polynomial ring?

In a polynomial ring

$$R[x],$$

the symbol x is a *formal indeterminate*. It is not automatically a number. A polynomial is a finite expression

$$a_0 + a_1x + \cdots + a_nx^n \quad (a_i \in R).$$

Thus in $\mathbb{Z}[x]$,

$$2x^2 + 3x + 5$$

is a polynomial with integer coefficients. Here x is only a formal symbol.

1.4.2 2. What is x^2 ?

The symbol

$$x^2$$

means $x \cdot x$, the square of the indeterminate x . It is itself a polynomial. For example:

$$x = 0 + 1 \cdot x, \quad x^2 = 0 + 0 \cdot x + 1 \cdot x^2.$$

1.4.3 3. Evaluation

Once one chooses an element $a \in R$, one may evaluate:

$$f(x) \mapsto f(a).$$

For example, if

$$f(x) = 2x^2 + 3x + 5 \in \mathbb{Z}[x],$$

then

$$f(4) = 2 \cdot 4^2 + 3 \cdot 4 + 5 = 49.$$

Thus x is formal in the polynomial ring, but becomes an actual value under evaluation.

1.4.4 4. The ring $\mathbb{Z}/n\mathbb{Z}$

The ring

$$\mathbb{Z}/n\mathbb{Z}$$

consists of residue classes modulo n :

$$\bar{0}, \bar{1}, \dots, \overline{n-1}.$$

Two integers define the same class if and only if they differ by a multiple of n :

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

Example in $\mathbb{Z}/6\mathbb{Z}$:

$$\bar{2} = \bar{8} = \overline{-4}.$$

Arithmetic is done modulo n .

1.4.5 5. The ring $\mathbb{Z}/p\mathbb{Z}$

If p is prime, then

$$\mathbb{Z}/p\mathbb{Z}$$

is a field, often denoted

$$\mathbb{F}_p.$$

For example, in $\mathbb{Z}/5\mathbb{Z}$,

$$\bar{2} + \bar{4} = \bar{1}, \quad \bar{2}\bar{3} = \bar{1}.$$

If n is not prime, then $\mathbb{Z}/n\mathbb{Z}$ usually has zero divisors. For example, in $\mathbb{Z}/6\mathbb{Z}$,

$$\bar{2} \cdot \bar{3} = \bar{0}.$$

1.4.6 6. Polynomial rings over $\mathbb{Z}/p\mathbb{Z}$

The ring

$$(\mathbb{Z}/5\mathbb{Z})[x]$$

consists of polynomials in x with coefficients modulo 5. For example:

$$\bar{3}x^2 + \bar{4}x + \bar{1}.$$

Addition and multiplication are performed as usual, with coefficients reduced modulo 5.

1.4.7 7. Quotient classes

If

$$A = R/I,$$

then for $f \in R$, its class in A is

$$\bar{f} = f + I.$$

Thus:

$$\bar{x} = x + I, \quad \overline{x^2} = x^2 + I.$$

Two elements define the same class if and only if their difference lies in I :

$$\bar{f} = \bar{g} \iff f - g \in I.$$

1.4.8 8. Example: $k[x]/(x^2)$

Let

$$A = k[x]/(x^2).$$

Then:

$$\bar{x} \neq 0$$

because $x \notin (x^2)$, but

$$\bar{x}^2 = 0$$

because $x^2 \in (x^2)$.

Every element of A has the form

$$a + b\bar{x}, \quad a, b \in k.$$

1.4.9 9. Example: $k[x]/(x(x-1))$

Let

$$A = k[x]/(x(x-1)).$$

Then in A ,

$$\bar{x}(\bar{x} - 1) = 0.$$

So $\bar{x}$ and $\bar{x} - 1$ are zero divisors.

The minimal primes are

$$(\bar{x}), \quad (\bar{x} - 1).$$

Hence

$$\text{Min}(A) = \{(\bar{x}), (\bar{x} - 1)\}.$$

Also

$$\text{Ass}(A) = \{(\bar{x}), (\bar{x} - 1)\}.$$

1.4.10 10. Example: $k[x, y]/(xy)$

Let

$$A = k[x, y]/(xy).$$

Then

$$\bar{x}\bar{y} = 0,$$

but $\bar{x} \neq 0$ and $\bar{y} \neq 0$.

The minimal primes are

$$(\bar{x}), \quad (\bar{y}),$$

so

$$\text{Min}(A) = \{(\bar{x}), (\bar{y})\}.$$

Also

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}),$$

hence

$$\text{Ass}(A) = \{(\bar{x}), (\bar{y})\}.$$

1.4.11 11. Example: $k[x, y]/(x^2, xy)$

Let

$$A = k[x, y]/(x^2, xy).$$

Then

$$\bar{x}^2 = 0, \quad \bar{x}\bar{y} = 0.$$

The radical of (x^2, xy) is (x) , so

$$\text{Min}(A) = \{(\bar{x})\}.$$

Now

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}),$$

so

$$(\bar{x}, \bar{y}) \in \text{Ass}(A).$$

Thus

$$\text{Ass}(A) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

Therefore

$$\text{Min}(A) \subsetneq \text{Ass}(A).$$

The prime $(\bar{x}, \bar{y})$ is an embedded associated prime.

1.4.12 12. Interpretation

Minimal primes describe the main irreducible components of the ring. Associated primes describe actual annihilators of elements, hence zero-divisor structure and possible embedded components.

Thus:

minimal primes = top geometry, associated primes = top geometry + hidden annihilation structure

1.5 Concrete meaning of x , x^2 , $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{Z}/p\mathbb{Z}$, quotient classes, minimal primes, and associated primes

1.5.1 1. Polynomial rings and formal indeterminates

Let R be a ring. The polynomial ring

$$R[x]$$

consists of all finite expressions

$$a_0 + a_1x + \cdots + a_nx^n, \quad a_i \in R.$$

Here x is a *formal indeterminate*. It is not automatically a number. It is simply a symbol used to build polynomials.

Thus:

$$x = 0 + 1 \cdot x, \quad x^2 = 0 + 0 \cdot x + 1 \cdot x^2.$$

So x^2 means the product $x \cdot x$, and it is itself a polynomial.

For example, in $\mathbb{Z}[x]$,

$$2x^2 + 3x + 5$$

is a polynomial with integer coefficients. At this stage, x is not evaluated at any integer.

1.5.2 2. Evaluation of polynomials

If $f(x) \in R[x]$ and $a \in R$, then we may evaluate f at a :

$$f(x) \mapsto f(a).$$

For example, if

$$f(x) = 2x^2 + 3x + 5 \in \mathbb{Z}[x],$$

then

$$f(4) = 2 \cdot 4^2 + 3 \cdot 4 + 5 = 32 + 12 + 5 = 49.$$

Thus there is a fundamental distinction:

- in $R[x]$, x is formal;
- after evaluation, x is replaced by an actual element of a ring.

1.5.3 3. The rings $\mathbb{Z}/n\mathbb{Z}$ and $\mathbb{Z}/p\mathbb{Z}$

The ring

$$\mathbb{Z}/n\mathbb{Z}$$

is the ring of integers modulo n . Its elements are residue classes

$$\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1},$$

where

$$\bar{a} = \bar{b} \iff a \equiv b \pmod{n} \iff n \mid (a - b).$$

Example: $\mathbb{Z}/6\mathbb{Z}$

The classes are

$$\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}.$$

We have

$$\bar{2} = \bar{8} = \overline{-4},$$

and arithmetic is done modulo 6:

$$\bar{4} + \bar{5} = \bar{9} = \bar{3}, \quad \bar{2} \cdot \bar{4} = \bar{8} = \bar{2}.$$

Also

$$\bar{2} \cdot \bar{3} = \bar{0},$$

so $\bar{2}$ and $\bar{3}$ are zero divisors.

Example: $\mathbb{Z}/5\mathbb{Z}$

Since 5 is prime, the ring

$$\mathbb{Z}/5\mathbb{Z}$$

is a field, often denoted $\mathbb{F}_5$. Its elements are

$$\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}.$$

Examples:

$$\bar{2} + \bar{4} = \bar{6} = \bar{1}, \quad \bar{2} \cdot \bar{3} = \bar{6} = \bar{1}.$$

Every nonzero element has a multiplicative inverse.

In general:

- $\mathbb{Z}/p\mathbb{Z}$ is a field if p is prime;
- $\mathbb{Z}/n\mathbb{Z}$ usually has zero divisors if n is composite.

1.5.4 4. Polynomial rings over $\mathbb{Z}/p\mathbb{Z}$

One may form polynomial rings over finite fields:

$$(\mathbb{Z}/p\mathbb{Z})[x].$$

For example, in $(\mathbb{Z}/5\mathbb{Z})[x]$,

$$f(x) = \bar{3}x^2 + \bar{4}x + \bar{1}$$

is a polynomial.

Addition is coefficientwise modulo 5. For example,

$$(\bar{3}x^2 + \bar{4}x + \bar{1}) + (\bar{4}x^2 + \bar{2}x + \bar{3}) = \bar{2}x^2 + \bar{x} + \bar{4},$$

because

$$\bar{3} + \bar{4} = \bar{2}, \quad \bar{4} + \bar{2} = \bar{1}, \quad \bar{1} + \bar{3} = \bar{4}.$$

1.5.5 5. Quotient rings and classes

Let R be a ring and $I \subseteq R$ an ideal. The quotient ring

$$R/I$$

consists of residue classes modulo I . For $f \in R$, its class is written

$$\bar{f} = f + I.$$

Thus:

$$\bar{x} = x + I, \quad \overline{x^2} = x^2 + I.$$

Two elements $f, g \in R$ define the same class if and only if

$$\bar{f} = \bar{g} \iff f - g \in I.$$

So $\bar{f}$ is not the original polynomial f , but its equivalence class modulo I .

1.5.6 6. Example: $k[x]/(x^2)$

Let

$$A = k[x]/(x^2),$$

where k is a field.

In this quotient ring, the class $\bar{x}$ satisfies

$$\bar{x}^2 = 0,$$

because $x^2 \in (x^2)$. However,

$$\bar{x} \neq 0,$$

because $x \notin (x^2)$.

Every element of A has the form

$$a + b\bar{x}, \quad a, b \in k.$$

Indeed, every polynomial

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

has class

$$\bar{f} = a_0 + a_1\bar{x},$$

since all terms of degree at least 2 vanish in the quotient.

Examples.

$$\overline{x + x^2} = \bar{x}, \quad \overline{1 + x^3} = 1, \quad (\bar{x})^2 = 0.$$

This is a standard example of a nonreduced ring: $\bar{x} \neq 0$, but $\bar{x}$ is nilpotent.

1.5.7 7. Example: $k[x]/(x(x-1))$

Let

$$A = k[x]/(x(x-1)).$$

Then in A ,

$$\bar{x}(\bar{x}-1) = 0.$$

Hence $\bar{x}$ and $\bar{x}-1$ are zero divisors.

This ring encodes the relation

$$x(x-1) = 0.$$

Geometrically, it corresponds to two points on the affine line: $x = 0$ and $x = 1$.

Examples.

$$\bar{x} \neq 0, \quad \bar{x}-1 \neq 0, \quad \bar{x}(\bar{x}-1) = 0.$$

1.5.8 8. Example: $k[x, y]/(xy)$

Let

$$A = k[x, y]/(xy).$$

Then

$$\bar{x}\bar{y} = 0,$$

but neither $\bar{x}$ nor $\bar{y}$ is zero. Indeed,

$$x \notin (xy), \quad y \notin (xy).$$

This is one of the most important examples:

$$\bar{x}\bar{y} = 0 \quad \text{with} \quad \bar{x} \neq 0, \quad \bar{y} \neq 0.$$

So $\bar{x}$ and $\bar{y}$ are nonzero zero divisors.

Examples of classes.

$$\overline{x+xy} = \bar{x}, \quad \overline{y+x^2y} = \bar{y}, \quad \overline{xy} = 0.$$

Annihilators. We compute:

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}).$$

Indeed, $\bar{f} \in \text{Ann}(\bar{x})$ means

$$\bar{f}\bar{x} = 0 \iff fx \in (xy),$$

which forces $f \in (y)$. Similarly for $\text{Ann}(\bar{y})$.

1.5.9 9. Example: $k[x, y]/(x^2, xy)$

Let

$$A = k[x, y]/(x^2, xy).$$

Then in A ,

$$\bar{x}^2 = 0, \quad \bar{x}\bar{y} = 0.$$

Thus $\bar{x}$ is killed by both $\bar{x}$ and $\bar{y}$.

We compute:

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}).$$

Indeed, if $f\bar{x} = 0$, then

$$fx \in (x^2, xy) = x(x, y),$$

so $f \in (x, y)$.

This example is very important because it has an embedded associated prime.

1.5.10 10. Minimal primes

Let $A = R/I$. A prime ideal $\mathfrak{p} \subseteq R$ is *minimal over I* if:

- $I \subseteq \mathfrak{p}$,
- $\mathfrak{p}$ is prime,
- there is no smaller prime ideal between I and $\mathfrak{p}$.

The corresponding primes in $A = R/I$ are called the *minimal primes* of A . We write

$$\text{Min}(A)$$

for the set of minimal primes.

Interpretation. Minimal primes describe the top-level irreducible components of the ring or affine variety.

Example: $k[x]$

Since $k[x]$ is a domain, the zero ideal is prime. Thus

$$\text{Min}(k[x]) = \{(0)\}.$$

Example: $k[x]/(x^2)$

The radical of (x^2) is (x) , so

$$\text{Min}(k[x]/(x^2)) = \{(\bar{x})\}.$$

Example: $k[x]/(x(x-1))$

The ideal $(x(x-1))$ has two minimal primes:

$$(x), \quad (x-1).$$

Hence

$$\text{Min}\left(k[x]/(x(x-1))\right) = \{(\bar{x}), (\bar{x}-1)\}.$$

Example: $k[x, y]/(xy)$

The ideal (xy) has two minimal primes:

$$(x), \quad (y).$$

Therefore

$$\text{Min}\left(k[x, y]/(xy)\right) = \{(\bar{x}), (\bar{y})\}.$$

Example: $k[x, y]/(x^2, xy)$

We have

$$\sqrt{(x^2, xy)} = (x),$$

so

$$\text{Min}\left(k[x, y]/(x^2, xy)\right) = \{(\bar{x})\}.$$

1.5.11 11. Associated primes

A prime ideal $\mathfrak{p} \subseteq A$ is an *associated prime* of the A -module A if

$$\mathfrak{p} = \text{Ann}_A(a)$$

for some $a \in A$.

We write

$$\text{Ass}(A)$$

for the set of associated primes.

Interpretation. Associated primes detect actual annihilators of elements. They describe zero-divisor structure and may reveal hidden embedded components.

Example: $k[x]/(x^2)$

We have

$$\text{Ann}(\bar{x}) = (\bar{x}),$$

so

$$\text{Ass}\left(k[x]/(x^2)\right) = \{(\bar{x})\}.$$

Example: $k[x]/(x(x-1))$

We have

$$\text{Ann}(\bar{x}) = (\bar{x} - 1), \quad \text{Ann}(\bar{x} - 1) = (\bar{x}).$$

Hence

$$\text{Ass}\left(k[x]/(x(x-1))\right) = \{(\bar{x}), (\bar{x} - 1)\}.$$

Example: $k[x, y]/(xy)$

We computed

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}),$$

so

$$\text{Ass}\left(k[x, y]/(xy)\right) = \{(\bar{x}), (\bar{y})\}.$$

Example: $k[x, y]/(x^2, xy)$

We have already seen that

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}).$$

Also the minimal prime $(\bar{x})$ is associated in this Noetherian case. Therefore

$$\text{Ass}\left(k[x, y]/(x^2, xy)\right) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

Thus

$$\text{Min}(A) = \{(\bar{x})\} \subsetneq \text{Ass}(A) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

The prime $(\bar{x}, \bar{y})$ is an *embedded associated prime*.

1.5.12 12. Comparison of minimal primes and associated primes

Minimal primes describe the irreducible components:

top-level geometry.

Associated primes describe annihilators of actual elements:

zero-divisor structure and embedded behavior.

In general, for Noetherian rings,

$$\text{Min}(A) \subseteq \text{Ass}(A),$$

and the inclusion may be strict.

1.5.13 13. Summary of the main examples

A	$\text{Min}(A)$	$\text{Ass}(A)$
$k[x]$	$\{(0)\}$	$\{(0)\}$
$k[x]/(x^2)$	$\{(\bar{x})\}$	$\{(\bar{x})\}$
$k[x]/(x(x-1))$	$\{(\bar{x}), (\bar{x}-1)\}$	$\{(\bar{x}), (\bar{x}-1)\}$
$k[x, y]/(xy)$	$\{(\bar{x}), (\bar{y})\}$	$\{(\bar{x}), (\bar{y})\}$
$k[x, y]/(x^2, xy)$	$\{(\bar{x})\}$	$\{(\bar{x}), (\bar{x}, \bar{y})\}$

1.5.14 14. Final intuition

The key points are:

- x, y are formal variables in polynomial rings;
- x^2, xy are formal products;
- quotient rings impose algebraic relations such as $x^2 = 0$ or $xy = 0$;
- $\bar{x}, \bar{y}$ are classes of x, y in the quotient;
- minimal primes describe the main geometric components;
- associated primes describe where actual annihilation occurs.

Thus one may think of:

minimal primes = geometry, associated primes = geometry plus hidden zero-divisor structure.

1.6 Concrete annihilators: $\mathbb{Z}/p\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$, polynomial rings, and quotient rings

1.6.1 1. Definition of annihilator

Let A be a ring and let $a \in A$. The annihilator of a is

$$\text{Ann}_A(a) = \{r \in A : ra = 0\}.$$

It is an ideal of A .

More generally, if $S \subseteq A$, then

$$\text{Ann}_A(S) = \{r \in A : rs = 0 \text{ for all } s \in S\}.$$

For a single element a , this is just $\text{Ann}_A(a)$.

Basic intuition. The annihilator of a measures which elements of the ring kill a by multiplication. So:

- if $\text{Ann}_A(a) = 0$, then no nonzero element kills a ;
- if $\text{Ann}_A(a) \neq 0$, then a is a zero divisor (unless $a = 0$);
- prime annihilators are exactly what lead to associated primes.

1.6.2 2. First trivial examples

For every ring A ,

$$\text{Ann}_A(0) = A,$$

because every element kills 0.

Also,

$$\text{Ann}_A(1) = 0,$$

because if $r \cdot 1 = 0$, then $r = 0$.

Thus in every nonzero ring:

$$\text{Ann}_A(0) = A, \quad \text{Ann}_A(1) = 0.$$

1.6.3 3. Annihilators in a domain

Suppose A is an integral domain. Then for every nonzero $a \in A$,

$$\text{Ann}_A(a) = 0.$$

Indeed, if $ra = 0$ and $a \neq 0$, then $r = 0$ because there are no zero divisors.

Examples. In $k[x]$, where k is a field,

$$\text{Ann}_{k[x]}(x) = 0, \quad \text{Ann}_{k[x]}(x^2 + 1) = 0, \quad \text{Ann}_{k[x]}(f) = 0$$

for every nonzero polynomial $f \in k[x]$.

So in a domain, annihilators of nonzero elements are always trivial.

1.6.4 4. Annihilators in the field $\mathbb{Z}/p\mathbb{Z}$

Let p be prime. Then

$$\mathbb{Z}/p\mathbb{Z}$$

is a field. Therefore every nonzero element is invertible, hence cannot have a nonzero annihilator.

So:

$$\text{Ann}_{\mathbb{Z}/p\mathbb{Z}}(\bar{0}) = \mathbb{Z}/p\mathbb{Z},$$

and for every $\bar{a} \neq \bar{0}$,

$$\text{Ann}_{\mathbb{Z}/p\mathbb{Z}}(\bar{a}) = 0.$$

Example: $\mathbb{Z}/5\mathbb{Z}$

The elements are

$$\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}.$$

Then:

$$\text{Ann}(\bar{0}) = \mathbb{Z}/5\mathbb{Z},$$

but

$$\text{Ann}(\bar{1}) = 0, \quad \text{Ann}(\bar{2}) = 0, \quad \text{Ann}(\bar{3}) = 0, \quad \text{Ann}(\bar{4}) = 0.$$

Indeed, for example, if $\bar{r}\bar{2} = \bar{0}$, then multiply by the inverse of $\bar{2}$, namely $\bar{3}$, to get $\bar{r} = \bar{0}$.

Conclusion. Fields have no nonzero zero divisors, and therefore all nonzero annihilators are trivial.

1.6.5 5. Annihilators in $\mathbb{Z}/n\mathbb{Z}$

Now let

$$A = \mathbb{Z}/n\mathbb{Z}.$$

This is much more interesting when n is composite.

For $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$,

$$\text{Ann}(\bar{a}) = \{\bar{r} \in \mathbb{Z}/n\mathbb{Z} : \bar{r}\bar{a} = \bar{0}\}.$$

This means

$$ra \equiv 0 \pmod{n},$$

or equivalently,

$$n \mid ra.$$

Let

$$d = \gcd(n, a).$$

Then one has the concrete formula

$$\text{Ann}_{\mathbb{Z}/n\mathbb{Z}}(\bar{a}) = \left(\frac{\bar{n}}{\bar{d}}\right).$$

Why? Write

$$n = dn_1, \quad a = da_1, \quad \gcd(n_1, a_1) = 1.$$

Then

$$n \mid ra \iff dn_1 \mid rda_1 \iff n_1 \mid ra_1.$$

Since $\gcd(n_1, a_1) = 1$, this is equivalent to

$$n_1 \mid r.$$

So r must be a multiple of n/d , exactly as claimed.

Example: $\mathbb{Z}/6\mathbb{Z}$

Annihilator of $\bar{2}$. Here

$$\gcd(6, 2) = 2, \quad \frac{6}{2} = 3.$$

So

$$\text{Ann}(\bar{2}) = (\bar{3}) = \{\bar{0}, \bar{3}\}.$$

Indeed,

$$\bar{0} \cdot \bar{2} = \bar{0}, \quad \bar{3} \cdot \bar{2} = \bar{6} = \bar{0}.$$

Annihilator of $\bar{3}$. Here

$$\gcd(6, 3) = 3, \quad \frac{6}{3} = 2.$$

So

$$\text{Ann}(\bar{3}) = (\bar{2}) = \{\bar{0}, \bar{2}, \bar{4}\}.$$

Indeed,

$$\bar{2} \cdot \bar{3} = \bar{0}, \quad \bar{4} \cdot \bar{3} = \bar{12} = \bar{0}.$$

Annihilator of $\bar{4}$. Since $\gcd(6, 4) = 2$, we get

$$\text{Ann}(\bar{4}) = (\bar{3}) = \{\bar{0}, \bar{3}\}.$$

Annihilator of $\bar{1}$. Since $\gcd(6, 1) = 1$,

$$\text{Ann}(\bar{1}) = (\bar{6}) = 0.$$

Thus the nonzero elements $\bar{2}, \bar{3}, \bar{4}$ have nontrivial annihilators, while $\bar{1}, \bar{5}$ do not.

Example: $\mathbb{Z}/12\mathbb{Z}$

Annihilator of $\bar{3}$.

$$\gcd(12, 3) = 3, \quad \frac{12}{3} = 4.$$

Hence

$$\text{Ann}(\bar{3}) = (\bar{4}) = \{\bar{0}, \bar{4}, \bar{8}\}.$$

Annihilator of $\bar{4}$.

$$\gcd(12, 4) = 4, \quad \frac{12}{4} = 3.$$

Hence

$$\text{Ann}(\bar{4}) = (\bar{3}) = \{\bar{0}, \bar{3}, \bar{6}, \bar{9}\}.$$

Annihilator of $\bar{6}$.

$$\gcd(12, 6) = 6, \quad \frac{12}{6} = 2.$$

Hence

$$\text{Ann}(\bar{6}) = (\bar{2}) = \{\bar{0}, \bar{2}, \bar{4}, \bar{6}, \bar{8}, \bar{10}\}.$$

So in $\mathbb{Z}/n\mathbb{Z}$, annihilators can be computed explicitly using $\gcd$.

Deep point. In $\mathbb{Z}/n\mathbb{Z}$,

$$\bar{a} \text{ is a zero divisor} \iff \gcd(n, a) \neq 1 \iff \text{Ann}(\bar{a}) \neq 0.$$

1.6.6 6. Annihilators in polynomial rings $k[x]$

Let k be a field. Since $k[x]$ is a domain, every nonzero polynomial has trivial annihilator:

$$\text{Ann}_{k[x]}(f) = 0 \quad \text{for all } f \neq 0.$$

Examples

$$\text{Ann}(x) = 0, \quad \text{Ann}(x^2) = 0, \quad \text{Ann}(x^2 + 1) = 0, \quad \text{Ann}((x - 1)(x + 2)) = 0.$$

So in ordinary polynomial rings over a field, annihilators do not become interesting until we pass to quotients.

1.6.7 7. Annihilators in $k[x]/(x^2)$

Let

$$A = k[x]/(x^2).$$

Then $\bar{x} \neq 0$, but

$$\bar{x}^2 = 0.$$

So $\bar{x}$ has a nontrivial annihilator.

Annihilator of $\bar{x}$. We compute

$$\text{Ann}_A(\bar{x}) = \{\bar{f} : \bar{f}\bar{x} = 0\}.$$

This means

$$fx \in (x^2).$$

So f must be divisible by x . Therefore

$$\text{Ann}_A(\bar{x}) = (\bar{x}).$$

General element. Every element of A has the form

$$a + b\bar{x}, \quad a, b \in k.$$

Now:

- if $a \neq 0$, then $a + b\bar{x}$ is a unit, so

$$\text{Ann}(a + b\bar{x}) = 0;$$

- if $a = 0$ and $b \neq 0$, then $a + b\bar{x} = b\bar{x}$, and

$$\text{Ann}(b\bar{x}) = (\bar{x}).$$

So the annihilators in this ring are:

$$\text{Ann}(0) = A, \quad \text{Ann}(\bar{x}) = (\bar{x}), \quad \text{Ann}(1) = 0, \quad \text{Ann}(1 + \bar{x}) = 0.$$

Interpretation. This ring has only one nonzero prime ideal, namely $(\bar{x})$, and $\bar{x}$ is exactly the nilpotent element killed by that prime.

1.6.8 8. Annihilators in $k[x]/(x(x-1))$

Let

$$A = k[x]/(x(x-1)).$$

Then

$$\bar{x}(\bar{x}-1) = 0.$$

Annihilator of $\bar{x}$. Since $(\bar{x}-1)\bar{x} = 0$, we have

$$(\bar{x}-1) \subseteq \text{Ann}(\bar{x}).$$

Conversely, if $f\bar{x} = 0$, then

$$fx \in (x(x-1)).$$

So fx is divisible by $x(x-1)$, hence f is divisible by $x-1$. Therefore

$$\text{Ann}(\bar{x}) = (\bar{x}-1).$$

Annihilator of $\bar{x}-1$. Similarly,

$$\text{Ann}(\bar{x}-1) = (\bar{x}).$$

Thus:

$$\text{Ann}(\bar{x}) = (\bar{x}-1), \quad \text{Ann}(\bar{x}-1) = (\bar{x}).$$

Interpretation. Each component kills the other one. This is the algebraic reflection of the decomposition into the two points $x = 0$ and $x = 1$.

1.6.9 9. Annihilators in $k[x, y]/(xy)$

Let

$$A = k[x, y]/(xy).$$

Then

$$\bar{x}\bar{y} = 0.$$

Annihilator of $\bar{x}$. We want

$$\bar{f}\bar{x} = 0 \iff fx \in (xy).$$

This means fx is divisible by xy , so f must be divisible by y . Hence

$$\text{Ann}(\bar{x}) = (\bar{y}).$$

Annihilator of $\bar{y}$. Similarly,

$$\text{Ann}(\bar{y}) = (\bar{x}).$$

Annihilator of $\bar{x} + \bar{y}$. Now ask whether $\bar{x} + \bar{y}$ has a nontrivial annihilator. Suppose

$$\bar{f}(\bar{x} + \bar{y}) = 0.$$

Then

$$f(x + y) \in (xy).$$

But $x + y$ is not divisible by x or by y , so the only way this product lies in (xy) is that $f \in (xy)$, i.e. $\bar{f} = 0$. Therefore

$$\text{Ann}(\bar{x} + \bar{y}) = 0.$$

So in this ring:

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}), \quad \text{Ann}(\bar{x} + \bar{y}) = 0.$$

Concrete lesson. Not every nonunit-looking element is a zero divisor. The elements $\bar{x}$ and $\bar{y}$ are killed by nonzero ideals, but $\bar{x} + \bar{y}$ is not.

1.6.10 10. Annihilators in $k[x, y]/(x^2, xy)$

Let

$$A = k[x, y]/(x^2, xy).$$

Then

$$\bar{x}^2 = 0, \quad \bar{x}\bar{y} = 0.$$

Annihilator of $\bar{x}$. Since both $\bar{x}$ and $\bar{y}$ kill $\bar{x}$, we have

$$(\bar{x}, \bar{y}) \subseteq \text{Ann}(\bar{x}).$$

Conversely, if $\bar{f}\bar{x} = 0$, then

$$fx \in (x^2, xy) = x(x, y).$$

So $f \in (x, y)$. Therefore

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}).$$

Annihilator of $\bar{y}$. Suppose $\bar{f}\bar{y} = 0$. Then

$$fy \in (x^2, xy).$$

Since x^2 and xy are both divisible by x , and fy contains no forced factor x , the only possibility is that $f \in (x)$ and the remaining factor still vanishes. In fact one checks that no nonzero class kills $\bar{y}$, so

$$\text{Ann}(\bar{y}) = 0.$$

Annihilator of $\bar{x} + \bar{y}$. Again, $\bar{y}$ cannot be killed nontrivially, so $\bar{x} + \bar{y}$ is not killed by all of $(\bar{x}, \bar{y})$. In fact

$$\text{Ann}(\bar{x} + \bar{y}) = 0.$$

Thus:

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}), \quad \text{Ann}(\bar{y}) = 0, \quad \text{Ann}(\bar{x} + \bar{y}) = 0.$$

Interpretation. The element $\bar{x}$ is very fragile: it is killed by the maximal ideal $(\bar{x}, \bar{y})$. But $\bar{y}$ behaves much more like an element in a domain.

1.6.11 11. Annihilators and associated primes

A prime ideal $\mathfrak{p} \subseteq A$ is associated if

$$\mathfrak{p} = \text{Ann}_A(a)$$

for some $a \in A$.

So annihilators are the direct source of associated primes.

Examples

In $k[x]/(x^2)$:

$$\text{Ann}(\bar{x}) = (\bar{x}),$$

so

$$\text{Ass}(k[x]/(x^2)) = \{(\bar{x})\}.$$

In $k[x]/(x(x-1))$:

$$\text{Ann}(\bar{x}) = (\bar{x} - 1), \quad \text{Ann}(\bar{x} - 1) = (\bar{x}),$$

hence

$$\text{Ass}(k[x]/(x(x-1))) = \{(\bar{x}), (\bar{x} - 1)\}.$$

In $k[x, y]/(xy)$:

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}),$$

hence

$$\text{Ass}(k[x, y]/(xy)) = \{(\bar{x}), (\bar{y})\}.$$

In $k[x, y]/(x^2, xy)$:

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}),$$

so $(\bar{x}, \bar{y})$ is associated. Also the minimal prime $(\bar{x})$ is associated in this Noetherian example. Therefore

$$\text{Ass}(k[x, y]/(x^2, xy)) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

1.6.12 12. Annihilators and minimal primes

Minimal primes describe the main geometric components. Associated primes come from annihilators and may include embedded primes.

Comparison table

A	sample element a	$\text{Ann}_A(a)$	relevant primes
$\mathbb{Z}/5\mathbb{Z}$	2	0	no nonzero annihilator
$\mathbb{Z}/6\mathbb{Z}$	$\bar{2}$	$(\bar{3})$	$(\bar{3})$
$k[x]/(x^2)$	$\bar{x}$	$(\bar{x})$	$(\bar{x})$
$k[x]/(x(x-1))$	$\bar{x}$	$(\bar{x}-1)$	$(\bar{x}), (\bar{x}-1)$
$k[x, y]/(xy)$	$\bar{x}$	$(\bar{y})$	$(\bar{x}), (\bar{y})$
$k[x, y]/(x^2, xy)$	$\bar{x}$	$(\bar{x}, \bar{y})$	$(\bar{x}), (\bar{x}, \bar{y})$

1.6.13 13. Final summary

The annihilator is one of the most concrete and useful constructions in commutative algebra:

$$\text{Ann}_A(a) = \{r \in A : ra = 0\}.$$

Its behavior is easy to understand in important examples:

- in a field such as $\mathbb{Z}/p\mathbb{Z}$, every nonzero annihilator is trivial;
- in $\mathbb{Z}/n\mathbb{Z}$, annihilators are computed by gcd:

$$\text{Ann}(\bar{a}) = \left(\frac{\overline{n}}{\gcd(n, a)} \right);$$

- in a domain such as $k[x]$, every nonzero annihilator is trivial;
- in quotient rings such as $k[x]/(x^2)$ or $k[x, y]/(xy)$, annihilators become nontrivial and reveal zero-divisor structure;
- prime annihilators are exactly the source of associated primes.

Thus annihilators form the bridge between concrete multiplication computations and the abstract prime-ideal structure of the ring.

1.7 Annihilators, minimal primes, and associated primes for modules: concrete examples

1.7.1 1. Annihilators for modules: the basic definitions

Let R be a ring and let M be an R -module.

For an element $m \in M$, the annihilator of m is

$$\text{Ann}_R(m) = \{r \in R : rm = 0\}.$$

This is an ideal of R .

The annihilator of the whole module M is

$$\text{Ann}_R(M) = \{r \in R : rM = 0\}.$$

Equivalently,

$$\text{Ann}_R(M) = \bigcap_{m \in M} \text{Ann}_R(m).$$

Interpretation.

- $\text{Ann}_R(m)$ tells us which scalars kill one specific element m .
- $\text{Ann}_R(M)$ tells us which scalars kill the entire module.

These two ideals are usually different.

1.7.2 2. Associated primes and minimal primes of a module

A prime ideal $\mathfrak{p} \subseteq R$ is an *associated prime* of the R -module M if

$$\mathfrak{p} = \text{Ann}_R(m)$$

for some $m \in M$.

We write

$$\text{Ass}_R(M)$$

for the set of associated primes of M .

If M is finitely generated over a Noetherian ring, one defines the minimal primes of M as the minimal primes in the support of M , equivalently the minimal primes over $\text{Ann}_R(M)$. We write

$$\text{Min}_R(M)$$

for this set.

Interpretation.

- $\text{Min}_R(M)$ describes the main geometric components seen by the module.
- $\text{Ass}_R(M)$ records prime annihilators of actual elements of the module.

In the Noetherian finitely generated setting,

$$\text{Min}_R(M) \subseteq \text{Ass}_R(M),$$

and the inclusion may be strict.

1.7.3 3. The cyclic module R/I

A very important case is the cyclic module

$$M = R/I.$$

Then the element $\bar{1} = 1 + I \in R/I$ satisfies

$$\text{Ann}_R(\bar{1}) = I.$$

More generally, for $\bar{f} = f + I \in R/I$,

$$\text{Ann}_R(\bar{f}) = \{r \in R : rf \in I\} = (I : f),$$

the colon ideal.

So cyclic modules give a very concrete way to compute annihilators.

1.7.4 4. First examples over $\mathbb{Z}$

Example 1: $M = \mathbb{Z}$ as a $\mathbb{Z}$ -module

Take $R = \mathbb{Z}$ and $M = \mathbb{Z}$.

For any nonzero integer m ,

$$\text{Ann}_{\mathbb{Z}}(m) = 0,$$

because if $rm = 0$ in $\mathbb{Z}$, then $r = 0$.

Thus

$$\text{Ann}_{\mathbb{Z}}(\mathbb{Z}) = 0.$$

Since (0) is prime in $\mathbb{Z}$, we get

$$\text{Ass}_{\mathbb{Z}}(\mathbb{Z}) = \{(0)\}, \quad \text{Min}_{\mathbb{Z}}(\mathbb{Z}) = \{(0)\}.$$

So the free module $\mathbb{Z}$ behaves like a torsion-free object.

Example 2: $M = \mathbb{Z}/12\mathbb{Z}$

Now let

$$M = \mathbb{Z}/12\mathbb{Z}.$$

As a $\mathbb{Z}$ -module, this is a cyclic module:

$$M = \mathbb{Z}/(12).$$

First,

$$\text{Ann}_{\mathbb{Z}}(M) = (12),$$

because 12 kills every class modulo 12, and no smaller ideal does.

Now compute annihilators of specific elements.

Element $\bar{1}$.

$$\text{Ann}(\bar{1}) = (12).$$

Element $\bar{2}$. We want all $r \in \mathbb{Z}$ such that

$$r\bar{2} = \bar{0} \text{ in } \mathbb{Z}/12\mathbb{Z}.$$

This means

$$12 \mid 2r,$$

so r must be divisible by 6. Hence

$$\text{Ann}(\bar{2}) = (6).$$

Element $\bar{3}$. We need

$$12 \mid 3r,$$

so r must be divisible by 4. Thus

$$\text{Ann}(\bar{3}) = (4).$$

Element $\bar{4}$. We need

$$12 \mid 4r,$$

so r must be divisible by 3. Hence

$$\text{Ann}(\bar{4}) = (3).$$

Element $\bar{6}$. We need

$$12 \mid 6r,$$

so r must be divisible by 2. Hence

$$\text{Ann}(\bar{6}) = (2).$$

So the prime annihilators that occur are

$$(2), \quad (3).$$

Therefore

$$\text{Ass}_{\mathbb{Z}}(\mathbb{Z}/12\mathbb{Z}) = \{(2), (3)\}.$$

Also the minimal primes over (12) are exactly

$$(2), \quad (3),$$

so

$$\text{Min}_{\mathbb{Z}}(\mathbb{Z}/12\mathbb{Z}) = \{(2), (3)\}.$$

General formula. For $M = \mathbb{Z}/n\mathbb{Z}$ and $\bar{a} \in M$,

$$\text{Ann}_{\mathbb{Z}}(\bar{a}) = \left(\frac{n}{\gcd(n, a)} \right).$$

1.7.5 5. A vector-space example

Let k be a field, and let $M = k^n$ be a k -vector space.

For every nonzero $v \in k^n$,

$$\text{Ann}_k(v) = 0,$$

because only $0 \in k$ kills a nonzero vector.

Thus

$$\text{Ann}_k(k^n) = 0, \quad \text{Ass}_k(k^n) = \{(0)\}, \quad \text{Min}_k(k^n) = \{(0)\}.$$

So over a field, modules behave very simply: nonzero elements have trivial annihilator.

1.7.6 6. Polynomial modules: $M = k[x]/(x^2)$

Let

$$R = k[x], \quad M = R/(x^2).$$

Then M is an R -module.

Since $M = R/I$ with $I = (x^2)$,

$$\text{Ann}_R(M) = (x^2).$$

Now compute annihilators of concrete elements.

Element $\bar{1}$.

$$\text{Ann}_R(\bar{1}) = (x^2).$$

Element $\bar{x}$. We need all $f \in k[x]$ such that

$$f\bar{x} = 0 \text{ in } R/(x^2).$$

This means

$$fx \in (x^2),$$

so f must be divisible by x . Hence

$$\text{Ann}_R(\bar{x}) = (x).$$

Element $\overline{1+x}$. Suppose

$$f(\overline{1+x}) = 0.$$

Then

$$f(1+x) \in (x^2).$$

But $1+x$ is not divisible by x , so the only way this can happen is $f \in (x^2)$. Therefore

$$\text{Ann}_R(\overline{1+x}) = (x^2).$$

Thus the prime annihilator that appears is

$$(x).$$

So

$$\text{Ass}_R(k[x]/(x^2)) = \{(x)\}.$$

Also the only minimal prime over (x^2) is

$$(x),$$

hence

$$\text{Min}_R(k[x]/(x^2)) = \{(x)\}.$$

1.7.7 7. Polynomial modules: $M = k[x]/(x(x-1))$

Let

$$R = k[x], \quad M = R/(x(x-1)).$$

Then

$$\text{Ann}_R(M) = (x(x-1)).$$

Now compute annihilators of elements.

Element $\bar{1}$.

$$\text{Ann}(\bar{1}) = (x(x-1)).$$

Element $\bar{x}$. We need

$$f\bar{x} = 0 \iff fx \in (x(x-1)).$$

So fx must be divisible by $x(x-1)$, hence f must be divisible by $x-1$. Therefore

$$\text{Ann}(\bar{x}) = (x-1).$$

Element $\overline{x-1}$. Similarly,

$$\text{Ann}(\overline{x-1}) = (x).$$

Thus the associated primes are

$$\text{Ass}_R(k[x]/(x(x-1))) = \{(x), (x-1)\}.$$

These are also the minimal primes over $(x(x-1))$, so

$$\text{Min}_R(k[x]/(x(x-1))) = \{(x), (x-1)\}.$$

Interpretation. The module sees two components, corresponding to the two points $x = 0$ and $x = 1$.

1.7.8 8. Polynomial modules: $M = k[x, y]/(xy)$

Let

$$R = k[x, y], \quad M = R/(xy).$$

Then

$$\text{Ann}_R(M) = (xy).$$

Now compute element annihilators.

Element $\bar{x}$. We need

$$f\bar{x} = 0 \iff fx \in (xy).$$

Thus f must be divisible by y , so

$$\text{Ann}(\bar{x}) = (y).$$

Element $\bar{y}$. Similarly,

$$\text{Ann}(\bar{y}) = (x).$$

Element $\overline{x+y}$. Suppose

$$f(\overline{x+y}) = 0.$$

Then

$$f(x+y) \in (xy).$$

Since $x+y$ is not divisible by x or by y , the only possibility is

$$f \in (xy).$$

Therefore

$$\text{Ann}(\overline{x+y}) = (xy).$$

So the prime annihilators that occur are

$$(x), \quad (y).$$

Hence

$$\text{Ass}_R(k[x, y]/(xy)) = \{(x), (y)\}.$$

These are also the minimal primes over (xy) , so

$$\text{Min}_R(k[x, y]/(xy)) = \{(x), (y)\}.$$

Interpretation. The element $\bar{x}$ is killed by the y -axis, and $\bar{y}$ is killed by the x -axis.

1.7.9 9. An embedded-prime module: $M = k[x, y]/(x^2, xy)$

Let

$$R = k[x, y], \quad M = R/(x^2, xy).$$

Then

$$\text{Ann}_R(M) = (x^2, xy).$$

Now compute annihilators of elements.

Element $\bar{1}$.

$$\text{Ann}(\bar{1}) = (x^2, xy).$$

Element $\bar{x}$. We need

$$f\bar{x} = 0 \iff fx \in (x^2, xy) = x(x, y).$$

This is equivalent to $f \in (x, y)$. Hence

$$\text{Ann}(\bar{x}) = (x, y).$$

Element $\bar{y}$. We need

$$f\bar{y} = 0 \iff fy \in (x^2, xy).$$

Since fy belongs to $(x^2, xy) = x(x, y)$, it must be divisible by x . Because x and y are independent variables, this forces $f \in (x)$. Conversely, any multiple of x kills $\bar{y}$, because $xy \in (x^2, xy)$. Therefore

$$\text{Ann}(\bar{y}) = (x).$$

Thus we get two prime annihilators:

$$(x), \quad (x, y).$$

Hence

$$\text{Ass}_R(k[x, y]/(x^2, xy)) = \{(x), (x, y)\}.$$

Now

$$\sqrt{(x^2, xy)} = (x),$$

so the only minimal prime over $\text{Ann}_R(M)$ is

$$(x).$$

Thus

$$\text{Min}_R(k[x, y]/(x^2, xy)) = \{(x)\}.$$

Therefore

$$\text{Min}_R(M) = \{(x)\} \subsetneq \text{Ass}_R(M) = \{(x), (x, y)\}.$$

The prime (x, y) is an embedded associated prime.

1.7.10 10. Direct sums of modules

Let

$$R = k[x, y], \quad M = R/(x) \oplus R/(y).$$

Then

$$\text{Ann}_R(M) = \text{Ann}_R(R/(x)) \cap \text{Ann}_R(R/(y)) = (x) \cap (y) = (xy).$$

Now consider the two elements

$$m_1 = (\bar{1}, 0), \quad m_2 = (0, \bar{1}).$$

Their annihilators are:

$$\text{Ann}_R(m_1) = (x), \quad \text{Ann}_R(m_2) = (y).$$

So

$$\text{Ass}_R(M) = \{(x), (y)\}.$$

And the minimal primes over (xy) are again

$$(x), \quad (y).$$

Hence

$$\text{Min}_R(M) = \{(x), (y)\}.$$

This is a very concrete example showing that associated primes of a direct sum are the union of the associated primes of the summands.

1.7.11 11. A free polynomial module

Let

$$R = k[x], \quad M = R^2.$$

Then no nonzero polynomial kills all of R^2 , so

$$\text{Ann}_R(M) = 0.$$

Also, for a nonzero vector such as

$$m = (1, x),$$

if $f(1, x) = (0, 0)$, then $f = 0$. Hence

$$\text{Ann}_R(m) = 0.$$

Since $R = k[x]$ is a domain, (0) is prime. Therefore

$$\text{Ass}_R(R^2) = \{(0)\}, \quad \text{Min}_R(R^2) = \{(0)\}.$$

So free modules over domains behave like torsion-free modules.

1.7.12 12. Summary table

R	M	$\text{Ann}_R(M)$	$\text{Ass}_R(M)$	$\text{Min}_R(M)$
$\mathbb{Z}$	$\mathbb{Z}$	(0)	$\{(0)\}$	$\{(0)\}$
$\mathbb{Z}$	$\mathbb{Z}/12\mathbb{Z}$	(12)	$\{(2), (3)\}$	$\{(2), (3)\}$
$k[x]$	$k[x]/(x^2)$	(x^2)	$\{(x)\}$	$\{(x)\}$
$k[x]$	$k[x]/(x(x-1))$	$(x(x-1))$	$\{(x), (x-1)\}$	$\{(x), (x-1)\}$
$k[x, y]$	$k[x, y]/(xy)$	(xy)	$\{(x), (y)\}$	$\{(x), (y)\}$
$k[x, y]$	$k[x, y]/(x^2, xy)$	(x^2, xy)	$\{(x), (x, y)\}$	$\{(x)\}$
$k[x, y]$	$R/(x) \oplus R/(y)$	(xy)	$\{(x), (y)\}$	$\{(x), (y)\}$
$k[x]$	R^2	(0)	$\{(0)\}$	$\{(0)\}$

1.7.13 13. Final interpretation

For modules, one must always distinguish:

$$\text{Ann}_R(m) \quad \text{for one element } m \in M,$$

from

$$\text{Ann}_R(M) \quad \text{for the whole module.}$$

The annihilator of the whole module need not be prime and usually describes the global defining equations of the module. The annihilator of a specific element may be much smaller, and when it is prime, it contributes to the associated primes.

Thus:

$$\text{minimal primes of } M = \text{main geometric support of } M,$$

while

$$\text{associated primes of } M = \text{prime annihilators of actual elements of } M.$$

That is why associated primes are finer than minimal primes: they can detect embedded structure that the global annihilator $\text{Ann}_R(M)$ does not show by itself.

1.8 Concrete residue classes in $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{Z}/p\mathbb{Z}$, $k[x]/(x)$, $A[t]/(t)$, and $k[x, y]/(xy)$

1.8.1 1. General principle: what is a residue class in R/I ?

Let R be a ring and $I \subseteq R$ an ideal. An element of the quotient ring

$$R/I$$

is a residue class

$$\bar{f} = f + I = \{f + h : h \in I\}.$$

Thus $\bar{f}$ is not just one element f , but the whole set of elements differing from f by something in I .

Two elements $f, g \in R$ define the same class if and only if

$$\bar{f} = \bar{g} \iff f - g \in I.$$

So the quotient R/I identifies all elements that differ by an element of I .

1.8.2 2. Residue classes in $\mathbb{Z}/n\mathbb{Z}$

The quotient

$$\mathbb{Z}/n\mathbb{Z}$$

consists of congruence classes modulo n . For an integer a ,

$$\bar{a} = a + n\mathbb{Z} = \{a + nk : k \in \mathbb{Z}\}.$$

So $\bar{a}$ is the set of all integers congruent to a modulo n .

Example: $\mathbb{Z}/5\mathbb{Z}$

We have

$$\bar{2} = 2 + 5\mathbb{Z} = \{\dots, -8, -3, 2, 7, 12, 17, \dots\}.$$

Also

$$\bar{2} = \bar{7} = \overline{-3},$$

because

$$7 - 2 = 5, \quad -3 - 2 = -5,$$

and both are divisible by 5.

Thus the elements of $\mathbb{Z}/5\mathbb{Z}$ are

$$\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}.$$

Example: $\mathbb{Z}/6\mathbb{Z}$

Similarly,

$$\bar{4} = 4 + 6\mathbb{Z} = \{\dots, -8, -2, 4, 10, 16, \dots\}.$$

Also

$$\bar{4} = \bar{10} = \overline{-2}.$$

Thus in $\mathbb{Z}/n\mathbb{Z}$, a residue class is simply a set of integers having the same remainder modulo n .

1.8.3 3. Residue classes in $\mathbb{Z}/p\mathbb{Z}$

If p is prime, then

$$\mathbb{Z}/p\mathbb{Z}$$

is a field. The residue classes are defined exactly as above:

$$\bar{a} = a + p\mathbb{Z}.$$

Example: $\mathbb{Z}/7\mathbb{Z}$

$$\bar{3} = 3 + 7\mathbb{Z} = \{\dots, -11, -4, 3, 10, 17, 24, \dots\}.$$

So the only difference between $\mathbb{Z}/p\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$ is not how classes are formed, but the algebraic structure afterward: when p is prime, every nonzero class has a multiplicative inverse.

1.8.4 4. Residue classes in $k[x]/(x)$

Now let k be a field. Consider

$$k[x]/(x).$$

An element is a class

$$\bar{f} = f(x) + (x).$$

Two polynomials $f(x)$ and $g(x)$ lie in the same class if and only if

$$f(x) - g(x) \in (x),$$

that is, if $f(x) - g(x)$ is divisible by x .

This means precisely that f and g have the same constant term.

Indeed, if

$$f(x) = a_0 + a_1x + \dots + a_nx^n,$$

then

$$f(x) - a_0 = x(a_1 + a_2x + \dots + a_nx^{n-1}) \in (x),$$

so

$$\bar{f} = \overline{a_0}.$$

Thus every class is determined by the constant term alone.

Class of 0

$$\bar{0} = (x) = \{xq(x) : q(x) \in k[x]\}.$$

So this class consists of all polynomials divisible by x :

$$x, \quad x^2, \quad 3x + 7x^2, \quad x(x^3 + 1), \dots$$

Class of 1

$$\bar{1} = 1 + (x) = \{1 + xq(x) : q(x) \in k[x]\}.$$

Examples:

$$1, \quad 1 + x, \quad 1 + 3x^2, \quad 1 + x + x^4, \dots$$

Class of $a \in k$

$$\bar{a} = a + (x) = \{a + xq(x) : q(x) \in k[x]\}.$$

Thus in $k[x]/(x)$, two polynomials are equal if and only if they have the same constant term. Therefore

$$k[x]/(x) \cong k.$$

1.8.5 5. Residue classes in $A[t]/(t)$

Let A be any commutative ring. Consider

$$A[t]/(t).$$

An element is a class

$$\bar{f} = f(t) + (t).$$

If

$$f(t) = a_0 + a_1t + \dots + a_nt^n,$$

then

$$f(t) - a_0 \in (t),$$

so

$$\bar{f} = \overline{a_0}.$$

Thus every class is again determined by the constant term.

Class of $a \in A$

$$\bar{a} = a + (t) = \{a + tq(t) : q(t) \in A[t]\}.$$

Class of 0

$$\bar{0} = (t) = \{tq(t) : q(t) \in A[t]\}.$$

So $A[t]/(t)$ behaves exactly like A , and in fact

$$A[t]/(t) \cong A.$$

Example

In $\mathbb{Z}[t]/(t)$,

$$\overline{7 + 3t + t^2} = \bar{7},$$

and the class $\bar{7}$ consists of all polynomials

$$7 + tq(t).$$

1.8.6 6. Residue classes in $k[x, y]/(x)$

Before treating $k[x, y]/(xy)$, it is useful to look at

$$k[x, y]/(x).$$

Here two polynomials are equivalent if and only if their difference is divisible by x .

So every polynomial becomes equal to what remains after setting $x = 0$. In other words, only the y -part survives.

If

$$f(x, y) = g(y) + xh(x, y),$$

then

$$\bar{f} = \overline{g(y)}.$$

Thus every class has a representative depending only on y , and

$$k[x, y]/(x) \cong k[y].$$

Example

$$x^2 + xy + y^3 + 2y + 5$$

has class

$$y^3 + 2y + 5.$$

So the class of $g(y)$ is

$$g(y) + (x) = \{g(y) + xh(x, y) : h(x, y) \in k[x, y]\}.$$

1.8.7 7. Residue classes in $k[x, y]/(xy)$

Now consider

$$B = k[x, y]/(xy).$$

An element is a class

$$\bar{f} = f(x, y) + (xy).$$

Two polynomials are in the same class if and only if their difference is divisible by xy :

$$\bar{f} = \bar{g} \iff f - g \in (xy).$$

This means that all mixed terms containing both x and y vanish in the quotient.

What survives modulo (xy) ?

Write

$$f(x, y) = \sum_{i, j \geq 0} a_{ij} x^i y^j.$$

If $i \geq 1$ and $j \geq 1$, then $x^i y^j$ is divisible by xy , so its class is zero in the quotient.

Therefore the only terms that survive are:

- the constant term,
- pure powers of x : $x, x^2, x^3, \dots$,
- pure powers of y : $y, y^2, y^3, \dots$

So every class has a representative of the form

$$a + p(x) + q(y),$$

where $a \in k$, $p(x) \in xk[x]$, and $q(y) \in yk[y]$.

Equivalently, the whole class is

$$a + p(x) + q(y) + (xy),$$

that is, all polynomials of the form

$$a + p(x) + q(y) + xy h(x, y)$$

with $h(x, y) \in k[x, y]$.

Class of 0

$$\bar{0} = (xy) = \{xy h(x, y) : h(x, y) \in k[x, y]\}.$$

Examples of polynomials in this class are:

$$xy, \quad x^2y, \quad xy^2, \quad 3xy + 7x^4y^3, \dots$$

All of these become zero in the quotient.

Class of $\bar{x}$

$$\bar{x} = x + (xy) = \{x + xy h(x, y) : h(x, y) \in k[x, y]\}.$$

Examples:

$$x, \quad x + xy, \quad x + 7x^2y, \quad x + x^5y^2, \dots$$

All these are equal to $\bar{x}$ in the quotient.

However,

$$x + x^2$$

is *not* in the same class as x , because

$$(x + x^2) - x = x^2,$$

and $x^2 \notin (xy)$.

So

$$\overline{x + x^2} \neq \bar{x}.$$

Class of $\bar{y}$

Similarly,

$$\bar{y} = y + (xy) = \{y + xy h(x, y) : h(x, y) \in k[x, y]\}.$$

Examples:

$$y, \quad y + xy, \quad y + 5xy^2, \quad y + x^3y, \dots$$

Class of $\overline{x + y^2 + 3}$

This class is

$$x + y^2 + 3 + (xy).$$

So it consists of all polynomials of the form

$$x + y^2 + 3 + xy h(x, y).$$

Examples in this class:

$$\begin{aligned} x + y^2 + 3, \\ x + y^2 + 3 + xy, \\ x + y^2 + 3 + 7x^2y^5, \\ x + y^2 + 3 + xy(x + y + 1). \end{aligned}$$

Class of $\overline{x^2 + y^3}$

This class is

$$x^2 + y^3 + (xy).$$

So it contains:

$$x^2 + y^3, \quad x^2 + y^3 + xy, \quad x^2 + y^3 + x^7y^2, \dots$$

But

$$x^2 + y^3 + x$$

is not in this class, because the difference is x , which is not divisible by xy .

1.8.8 8. What are $\bar{x}$ and $\bar{y}$ concretely in $k[x, y]/(xy)$?

They are the residue classes

$$\bar{x} = x + (xy), \quad \bar{y} = y + (xy).$$

So:

- $\bar{x}$ is the class of all polynomials congruent to x modulo (xy) ,
- $\bar{y}$ is the class of all polynomials congruent to y modulo (xy) .

They are not zero, because

$$x \notin (xy), \quad y \notin (xy).$$

But their product is zero:

$$\bar{x}\bar{y} = \overline{xy} = 0.$$

Thus $\bar{x}$ and $\bar{y}$ are nonzero zero divisors.

Also,

$$\bar{x}^2 \neq 0, \quad \bar{y}^2 \neq 0,$$

because

$$x^2 \notin (xy), \quad y^2 \notin (xy).$$

So only mixed terms vanish, not pure powers of x or y .

1.8.9 9. Comparison of the main examples

The quotient always identifies elements that differ by something in the defining ideal:

- in $\mathbb{Z}/n\mathbb{Z}$, one adds multiples of n :

$$\bar{a} = a + n\mathbb{Z};$$

- in $k[x]/(x)$, one adds multiples of x :

$$\bar{a} = a + (x) = \{a + xq(x)\};$$

- in $A[t]/(t)$, one adds multiples of t :

$$\bar{a} = a + (t) = \{a + tq(t)\};$$

- in $k[x, y]/(x)$, one adds multiples of x :

$$g(y) + (x) = \{g(y) + xh(x, y)\};$$

- in $k[x, y]/(xy)$, one adds multiples of xy :

$$a + p(x) + q(y) + (xy) = \{a + p(x) + q(y) + xy h(x, y)\}.$$

Thus:

- in $k[x]/(x)$, the class is determined only by the constant term;
- in $A[t]/(t)$, likewise only the constant term survives;
- in $k[x, y]/(x)$, only the y -part survives;
- in $k[x, y]/(xy)$, the constant term, pure x -terms, and pure y -terms survive, while mixed xy -terms vanish.

1.8.10 10. Final summary

A residue class in any quotient R/I is always

$$f + I = \{f + h : h \in I\}.$$

So:

- in $\mathbb{Z}/n\mathbb{Z}$, two integers are equal if they differ by a multiple of n ;
- in $k[x]/(x)$, two polynomials are equal if they have the same constant term;
- in $A[t]/(t)$, two polynomials are equal if they have the same constant term;
- in $k[x, y]/(x)$, two polynomials are equal if they become equal after setting $x = 0$;
- in $k[x, y]/(xy)$, two polynomials are equal if they differ by a multiple of xy , so mixed terms vanish.

In particular,

$$\bar{x} = 0 \text{ in } k[x]/(x),$$

because $x \in (x)$, but

$$\bar{x} \neq 0 \text{ in } k[x, y]/(xy),$$

because $x \notin (xy)$.